

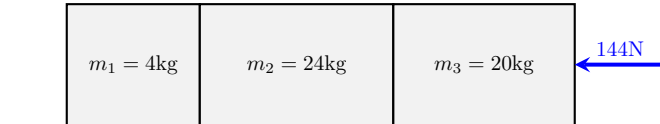
Dynamics Test

Ethan Wang

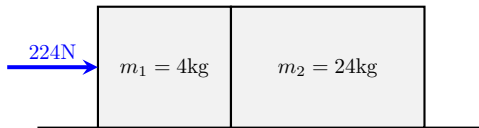
April 28, 2026

Blocks(60 points)

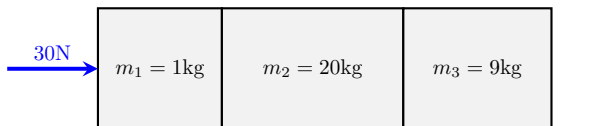
1. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 144N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



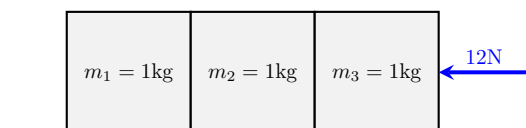
- (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
2. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 224N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



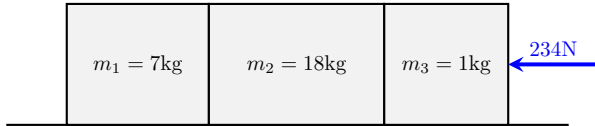
- (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
3. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 30N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



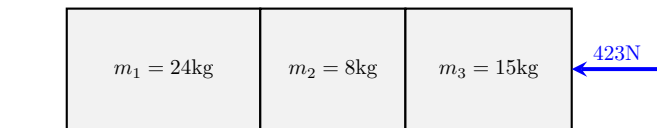
- (a) The acceleration of the system
 - (b) The acceleration of each block
 - (c) The net force on each block
 - (d) The contact forces for each block
4. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 12N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



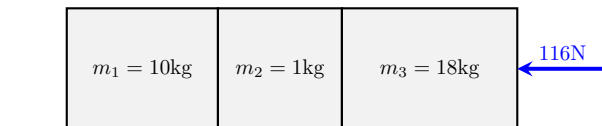
- (a) The acceleration of the system
 (b) The acceleration of each block
 (c) The net force on each block
 (d) The contact forces for each block
5. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 234N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



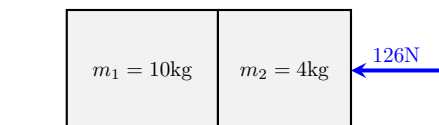
- (a) The acceleration of the system
 (b) The acceleration of each block
 (c) The net force on each block
 (d) The contact forces for each block
6. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 423N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system
 (b) The acceleration of each block
 (c) The net force on each block
 (d) The contact forces for each block
7. (6 points) The diagram shows 3 blocks in contact on a smooth surface. A 116N force acts on block 3 from the right, and the surface can be considered frictionless. Calculate:



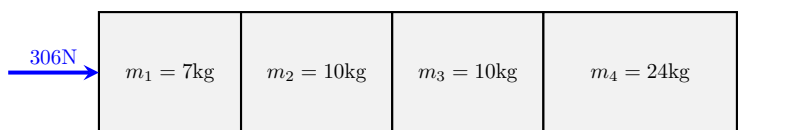
- (a) The acceleration of the system
 (b) The acceleration of each block
 (c) The net force on each block
 (d) The contact forces for each block
8. (6 points) The diagram shows 2 blocks in contact on a smooth surface. A 126N force acts on block 2 from the right, and the surface can be considered frictionless. Calculate:



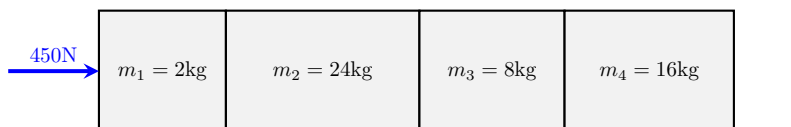
- (a) The acceleration of the system
 (b) The acceleration of each block
 (c) The net force on each block

(d) The contact forces for each block

9. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 306N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system
(b) The acceleration of each block
(c) The net force on each block
(d) The contact forces for each block
10. (6 points) The diagram shows 4 blocks in contact on a smooth surface. A 450N force acts on block 1 from the left, and the surface can be considered frictionless. Calculate:



- (a) The acceleration of the system
(b) The acceleration of each block
(c) The net force on each block
(d) The contact forces for each block